

CCT360 Project 1 – Process Documentation

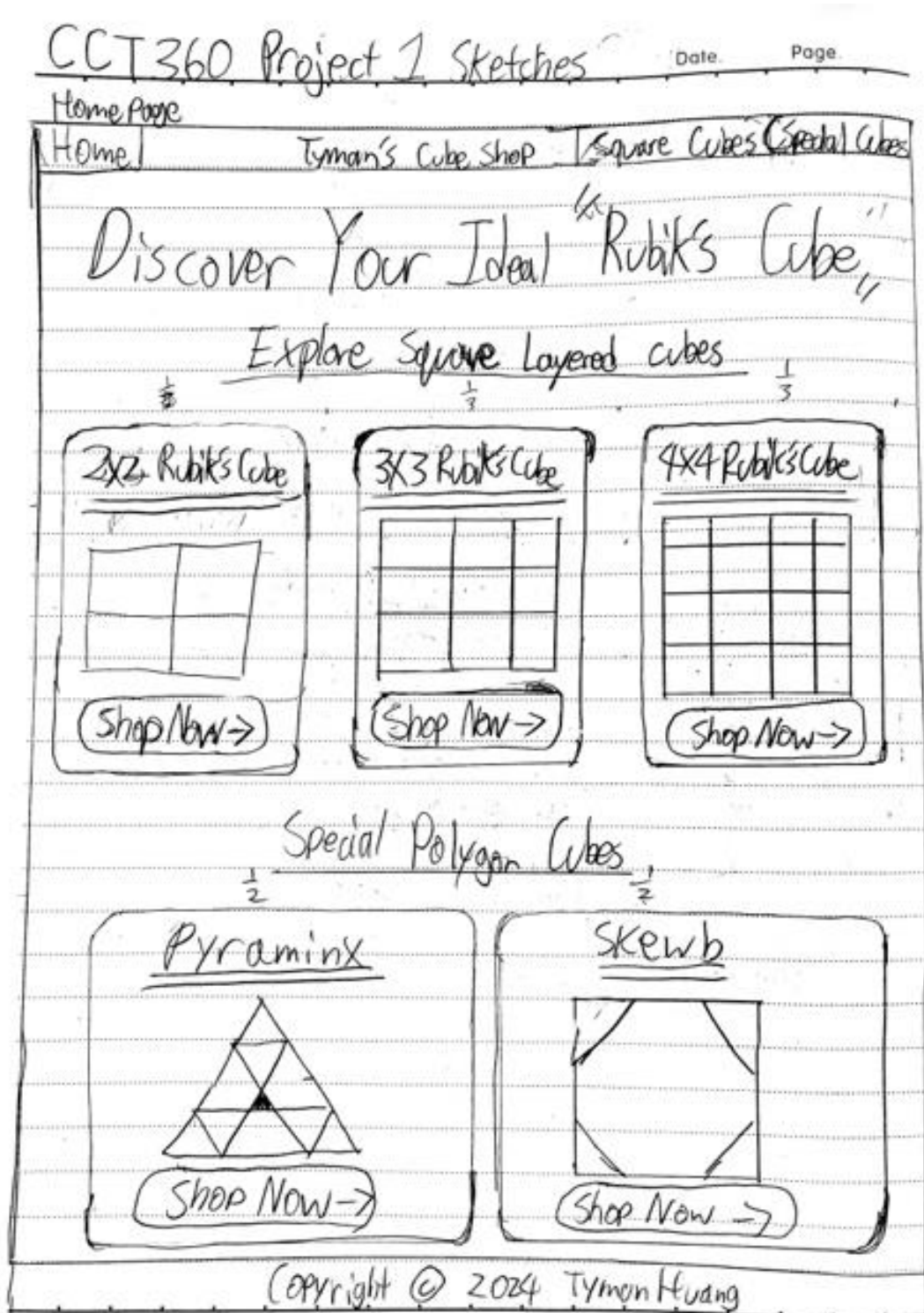
Action Plan Breakdown

1. Initially, I looked for items in my bedroom that appeared relatively new and could be used for a hypothetical online shopping site.
 - a. I chose five different types of Rubik's Cube as the products I will simulate on a shopping website.
2. I used my notebook and pencil to design my website in many variations and drafted many possible structures and page navigation orders.
3. My preliminary website layout originates from pencil-sketched designs (See preliminary home, product, and order page sketches):
 - a. Homepage
 - i. Navigation bar
 - ii. Title
 - iii. Heading 1
 - iv. Three product cards
 - v. Title 2
 - vi. Heading 2
 - vii. Two product cards
 - viii. Footer as a Copyright Statement
 - b. Five product pages (Each page has the same layout but different product texts and images)
 - i. Navigation bar
 - ii. Product title as heading 1
 - iii. Left horizontal column (1/2 fraction split)
 1. Image gallery of a product (6 images with price tag and left/right buttons)
 - iv. Right horizontal column (1/2 fraction split)
 1. Specifications as heading 2
 2. Unordered list as body 1
 3. Buy Now button
 - v. Footer as a Copyright Statement
 - c. Five checkout pages (Clicking "buy now" for each product will lead to the same order checkout page, but only the product title, image, and cost will differ as the form filing component is identical)
 - i. Navigation bar

- ii. Order checkout form as heading 1
 - iii. Left horizontal column (1/2 fraction split)
 - 1. Product title
 - 2. Product image
 - 3. Product price
 - iv. Right horizontal column (1/2 fraction split)
 - 1. Contact form
 - 2. Credit or Debit card form
 - 3. Order button
 - v. Footer as a Copyright Statement
- 4. After many revisions, my finalized webpage design has the following components:
 - a. Homepage
 - i. Navigation bar
 - 1. Home button to refresh the same page or go to the homepage from the product and order page
 - 2. Header title as shop name
 - 3. Products button to navigate to a page with all products listed as buttons that open each store item
 - ii. Title: **Discover Your Ideal Rubik's Cube**
 - iii. Subtitle 1: **Explore Square Layered Cubes**
 - iv. Three-column one row product cards
 - 1. Titles
 - 2. Product Images
 - 3. Shop now buttons
 - v. Subtitle 2: **Special Polygon Cubes**
 - vi. Two-column one row product cards
 - 1. Titles
 - 2. Product Images
 - 3. Shop now buttons
 - vii. Footer as a Copyright caption with my name
 - b. Product List page
 - i. Navigation bar (same as homepage)
 - ii. Title: All Rubik's Cube Products
 - iii. Subtitle 1: **Square Layered Cubes**
 - 1. Six product buttons (two rows and three columns)
 - iv. Subtitle 2: **Special Polygon Cubes**
 - 1. Three product buttons (one row and three columns)
 - c. 3x3 Rubik's Cube product details page

- i. Navigation bar (same as homepage)
- ii. Product title: **GAN 13 Magnetic Rubik's Cube (Heading 1)**
- iii. Left horizontal column (1/2 fraction split)
 - 1. Product image
- iv. Right horizontal column (1/2 fraction split)
 - 1. Specifications (Heading 2)
 - 2. Unordered list
 - a. Optimized for official Rubik's Cube competitions worldwide
 - b. Magnetic design allows smoother turns
 - c. Storage box, mini screwdriver, and tutorial manual sheet included
 - 3. Buy Now button and price label (\$20)
- v. Footer as a Copyright Statement

Preliminary Homepage Sketch



Preliminary Product and Order Page Sketch

Date: _____ Page: _____

Product Page layout - 3x3 Rubik's Cube as Example (5 different pages)

[Home](#)
[Tyman's Cube Shop](#)
[Square Cubes](#)
[Special Cubes](#)

GAN 13 Magnetic Rubik's Cube



\$20
USD

Specifications

- 3x3 layered cube for competitions (structure)
- Smoother turns with magnetic design
- Storage box and tutorial guide included (components)

Buy Now \$

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Order Page layout - 3x3 Rubik's Cube as Example (5 different pages)

[Home](#)
[Tyman's Cube Shop](#)
[Square Cubes](#)
[Special Cubes](#)

Order Checkout Form

GAN 13 Magnetic Rubik's Cube



First Name:

Last Name:

Address:

City: Postal Code: Province:

Credit & Debit Card ONLY

Card Number:

Expiry Date (mm/yy): CVV code:

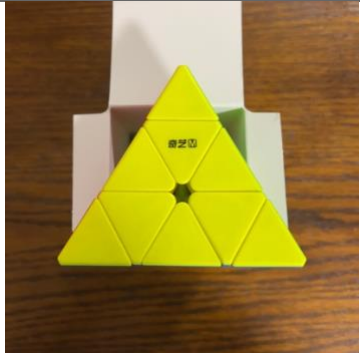
Total: USD \$20

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[back to home page](#)

5. Table 1: Image selection for each puzzle product's cover photo

Puzzle Type	Cover Photo (Front Face)	Rationale and Photography Practice Implemented
3x3 Classic Rubik's Cube		A top to bottom bird's eye view of a rubik's cube where a 1:1 square aspect ratio is used between all image assets, photographed with RAW camera filter from iPhone 13 Pro Max.
2x2 layer cube		The composition of all images in the folder is centered and symmetrical. There is a minimal depth of field, where the puzzles are sharply focused with a blurry wooden background.
4x4 layer cube		A darker shadow on the left half side of the wooden surface behind each cube image is intentionally photographed for a more artistic appeal.

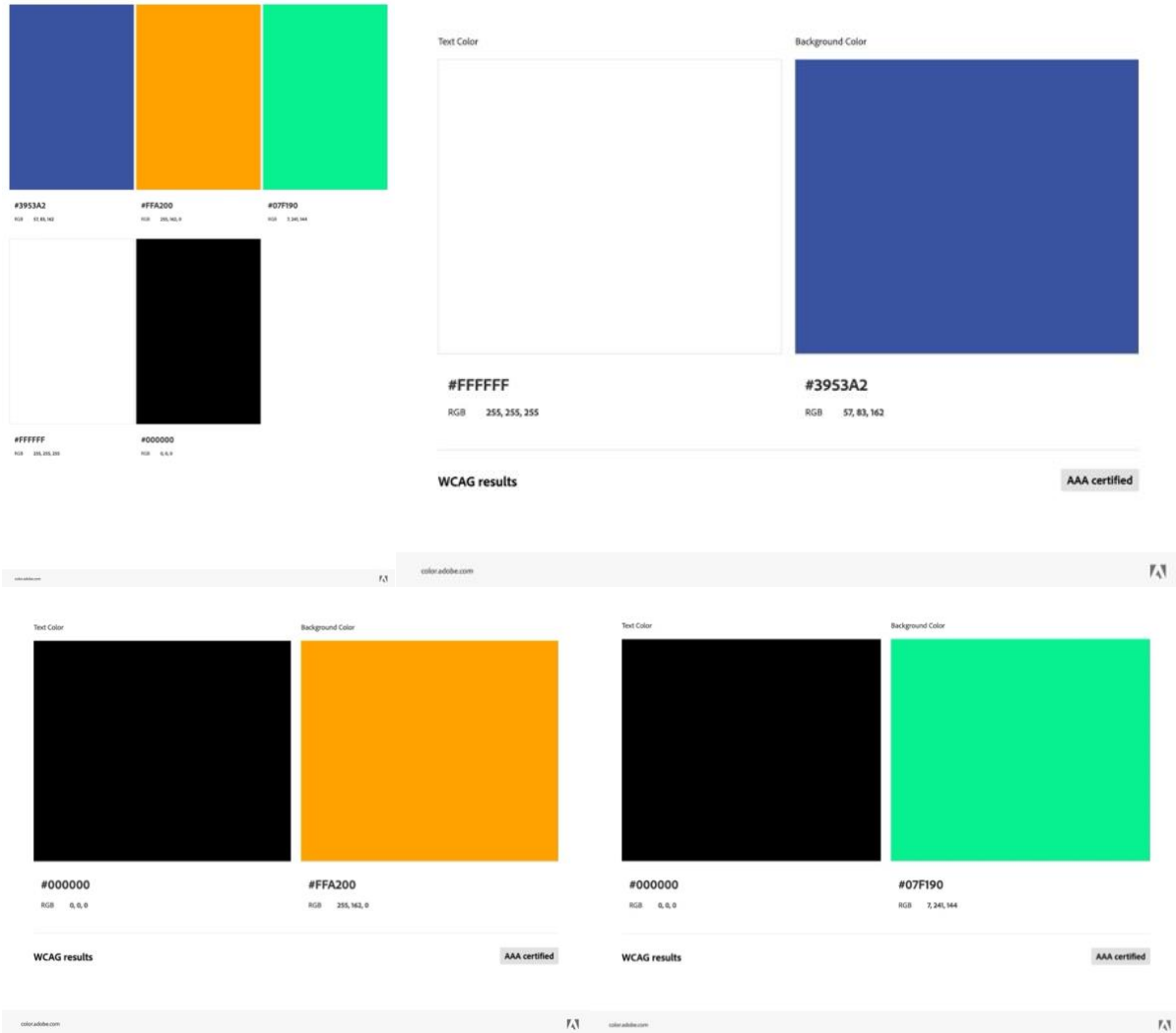
Pyraminx		The pyraminx is photographed on top of a small open rectangular container, so that the surface can be captured flatly when through an overhead shot.
Skewb		The skewb cube presents a different shadow pattern on the wooden surface, as the location and angle of gaps are different from regular layered cubes.

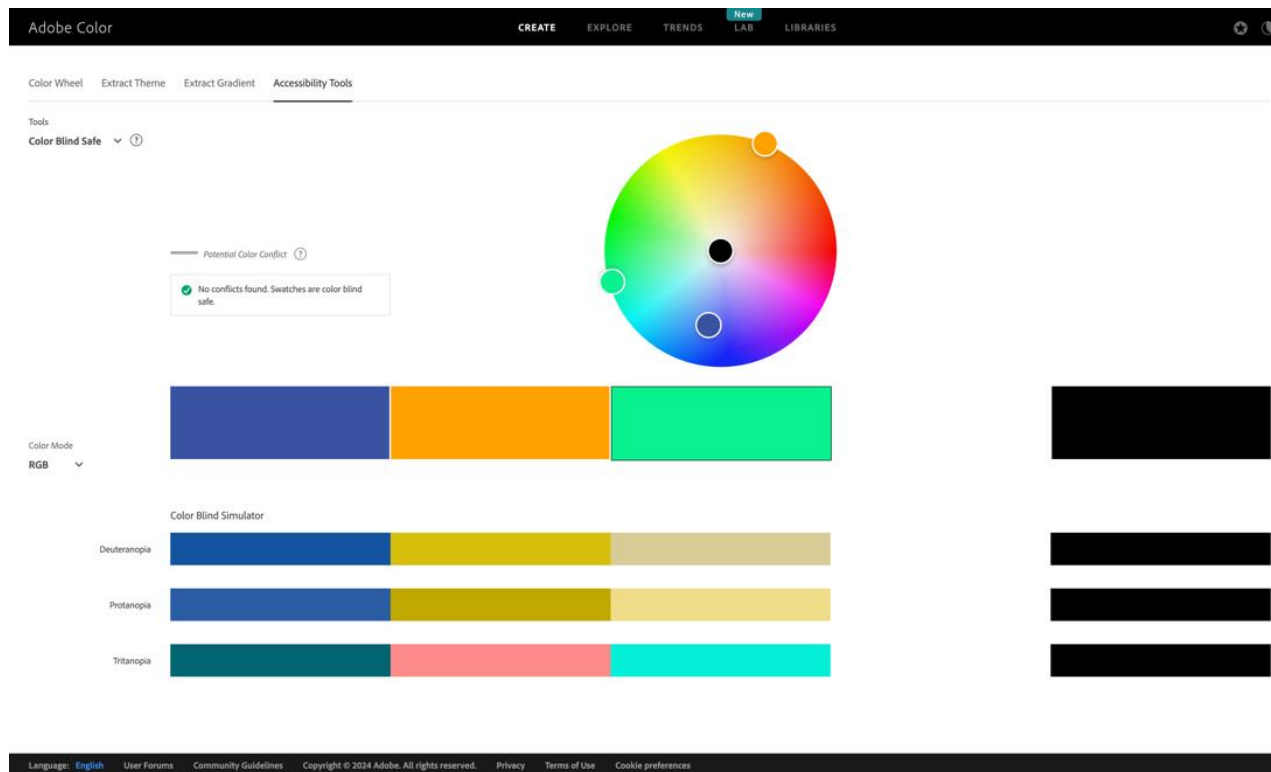
6. Puzzle product specifications writing

- a. GAN 13 Magnetic Rubik's cube
 - i. Optimized for official Rubik's Cube competitions worldwide
 - ii. Magnetic design allows smoother turns
 - iii. Storage box, mini screwdriver, and tutorial manual sheet included
- b. Moyu 2x2 Layered Rubik's cube
 - i. Simple starter twisty puzzle to help prepare for Rubik's cube
 - ii. Sturdy plastic tension framework to prevent corner twisting
 - iii. Mini screwdriver and tutorial manual sheet included
- c. Moyu 4x4 Layered Rubik's cube
 - i. Advance your skills beyond the 3x3 Rubik's Cube
 - ii. Stickerless design for better durability and aesthetics
 - iii. Mini screwdriver and tutorial manual sheet included
- d. Pyraminx
 - i. Four-faced triangular puzzle used in competitions
 - ii. Help improve geometry problem-solving skills
 - iii. Clear plastic case and tutorial manual sheet included
- e. Skewb
 - i. Diagonally oriented cube used in competitions
 - ii. Cornered rotation constraints in place for mind exercise
 - iii. Mini screwdriver and tutorial manual sheet included

Color Palette

Different colors are tested using WCAG 2.0 Web Accessibility standards to optimize color contrast and make it inclusive to website users with color-related visual impairments.





Project File Folder Structure

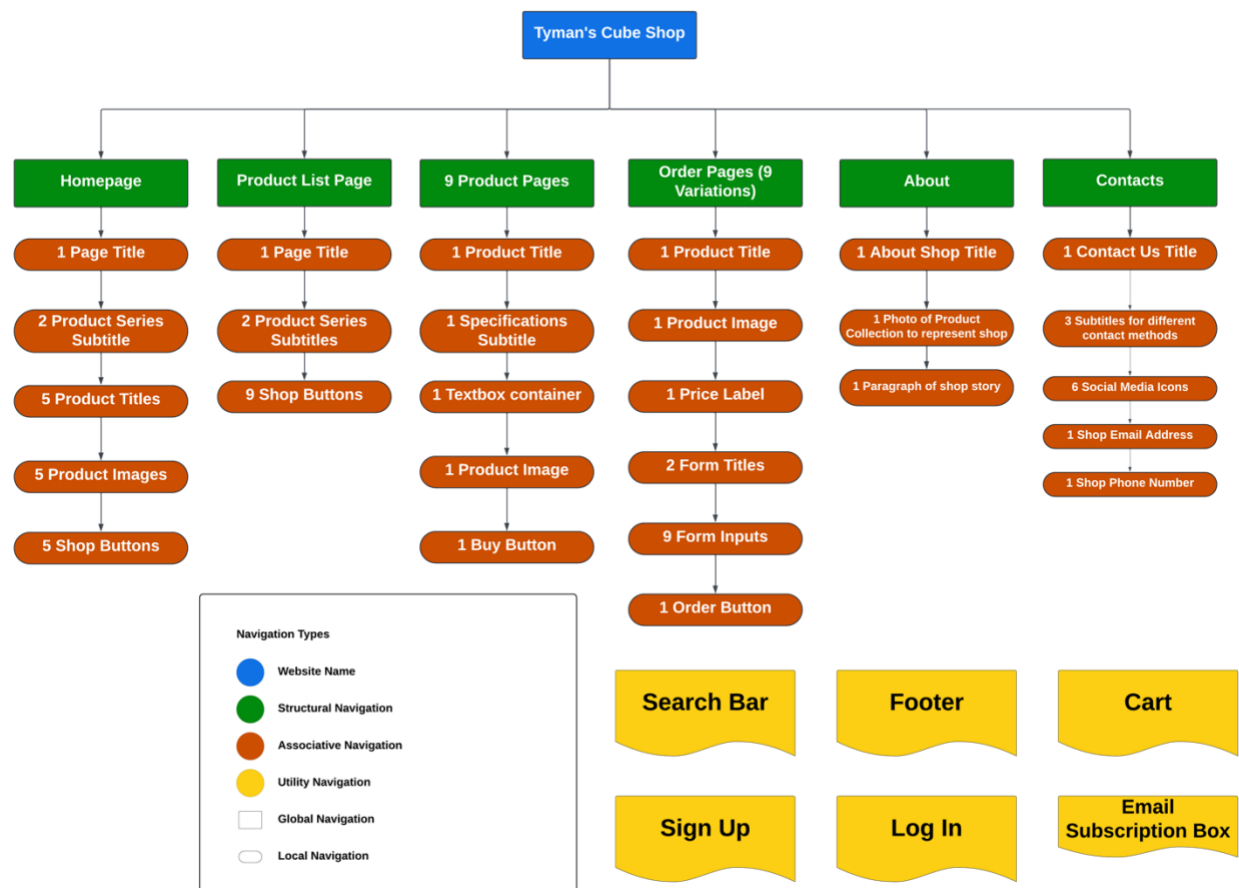
Master Folder: cct360_project1_tyman

- Process documentation write-up in PDF format
- CSS Folder
 - homepage.css
 - product_list.css
 - product_page.css
- Documentation files
- Media Assets
 - 2x2 cover.jpg
 - 3x3 cover.jpg
 - 4x4 cover.jpg
 - Pyraminx cover.jpg
 - Skewb cover.jpg
- homepage.html
- product_list.html
- product_page.html

Information Architecture (IA)

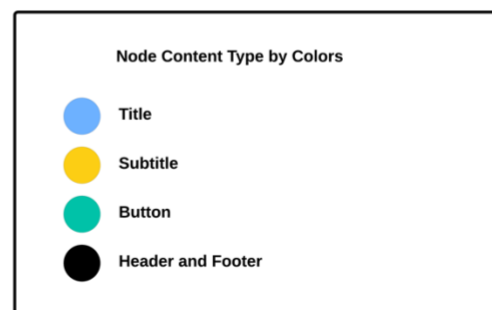
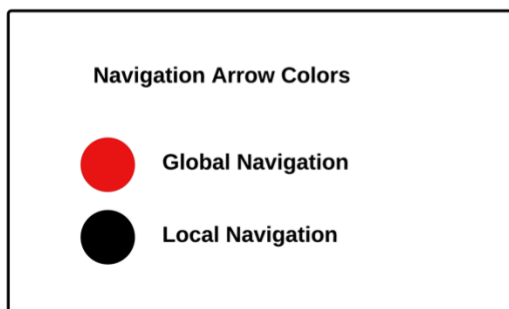
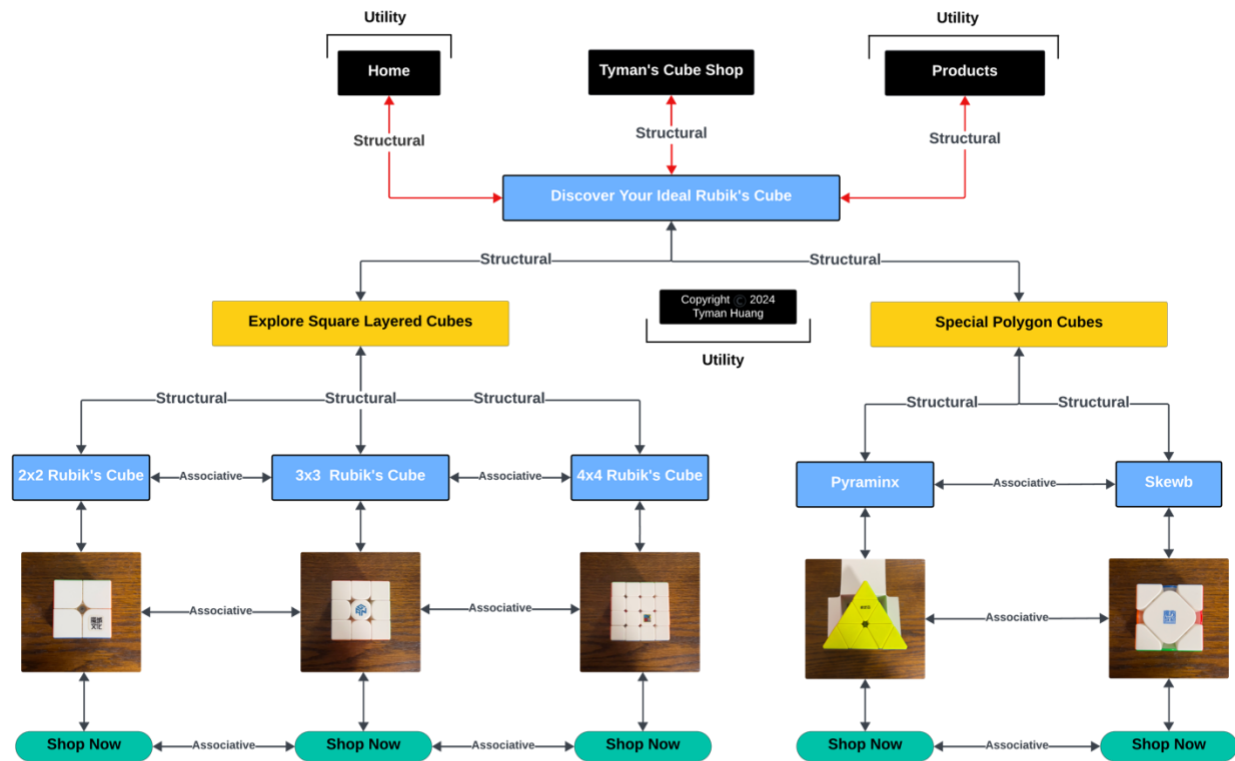
For the information architecture, I initially produced one sitemap flowchart to illustrate the connection between planned pages (not all implemented in the final website), where a legend table helps web designers understand the structural, associative, and utility navigation components across my Rubik's cube online shop containing all the necessary idealized web pages. Global and local navigation areas are also shape-differentiated, and a legend table is provided.

Information Architecture (IA) Master Sitemap

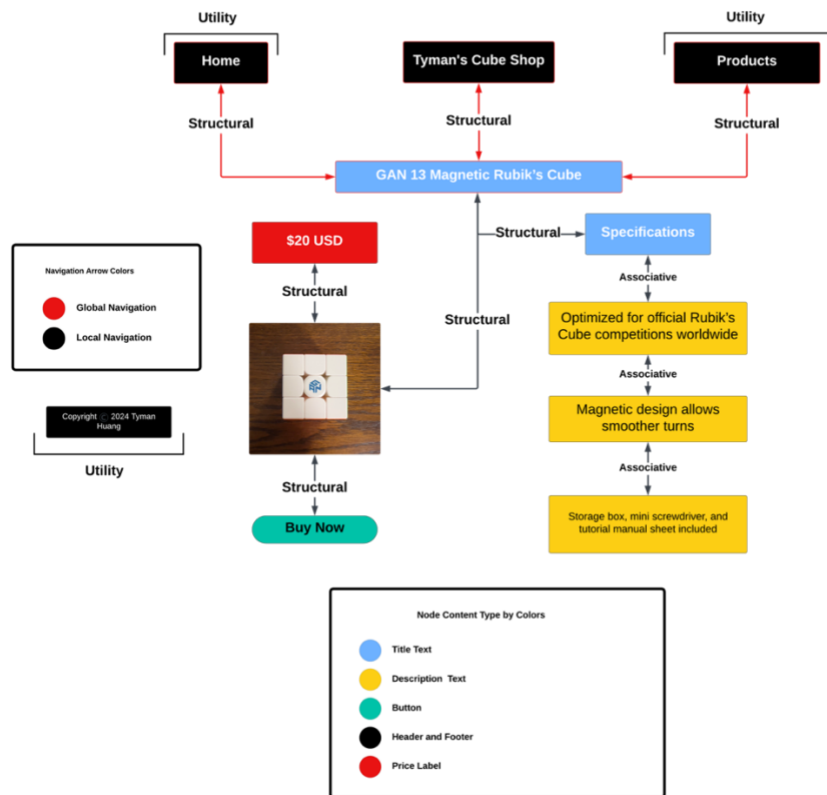


Afterward, I produced flowcharts of how each element within each page of my shop relates to one another. Structural, associative, and utility navigation are labeled in the arrows between each element (depicted as flowchart nodes). Global and local navigation are visualized through different arrow colors. The type of webpage element (title text, body text, button, image, etc.) is color-coded, corresponding to a concise legend table.

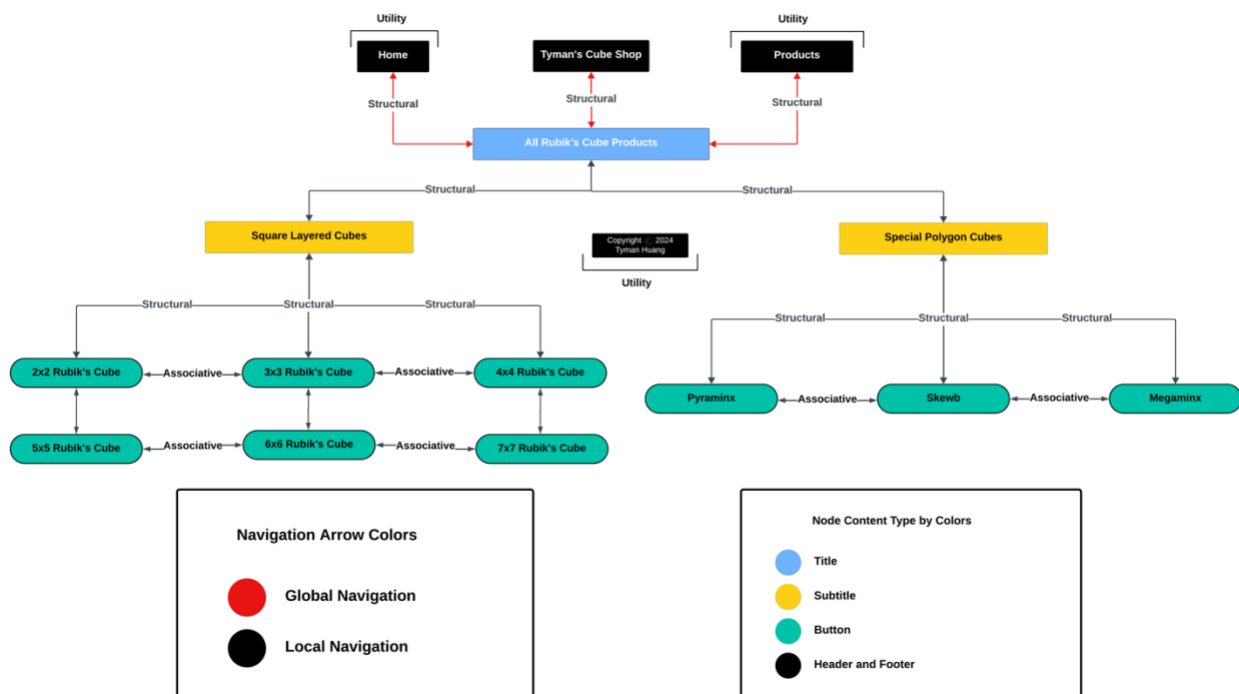
Information Architecture (IA) for Homepage



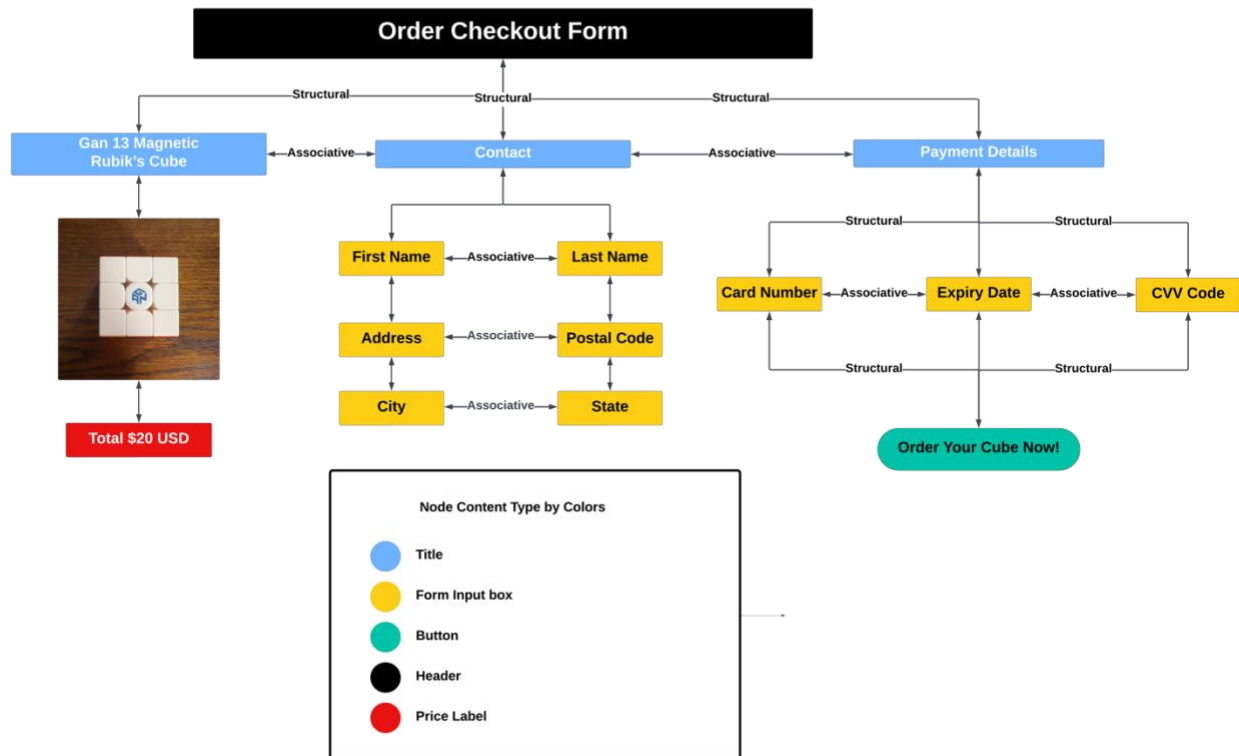
Information Architecture (IA) for Product Page (3x3 Rubik's Cube Example)



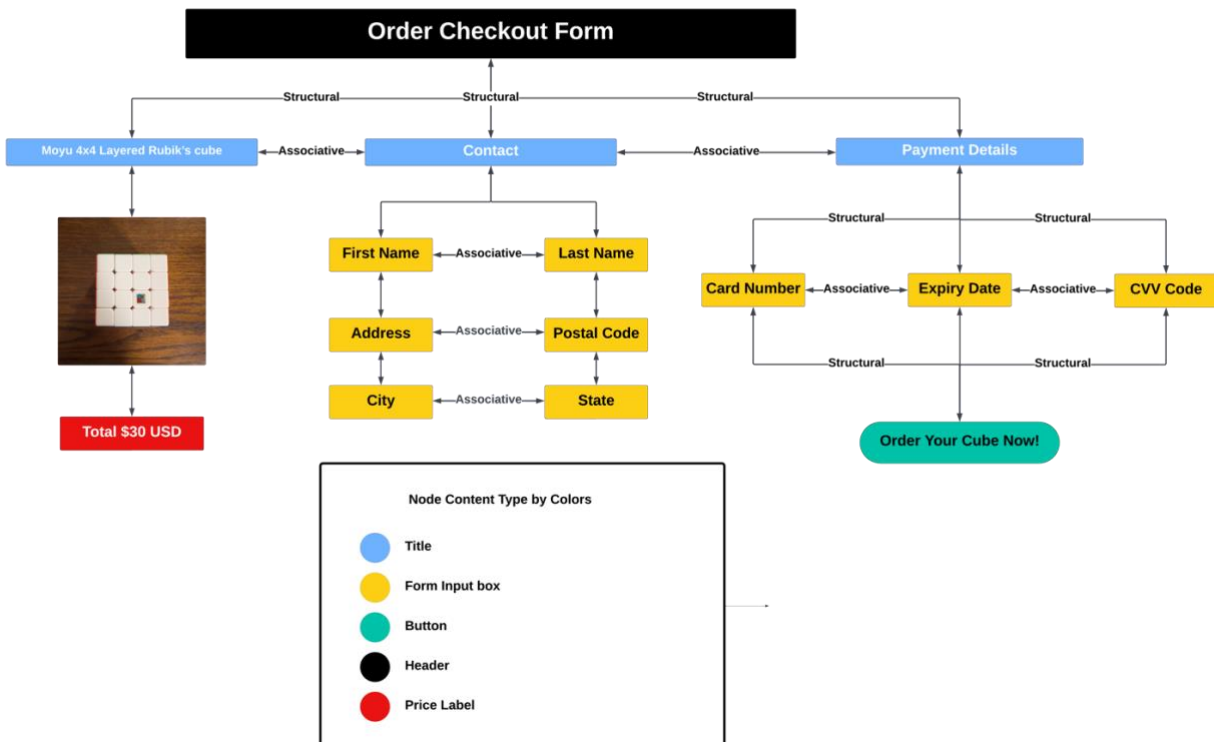
Information Architecture (IA) for Products List Page



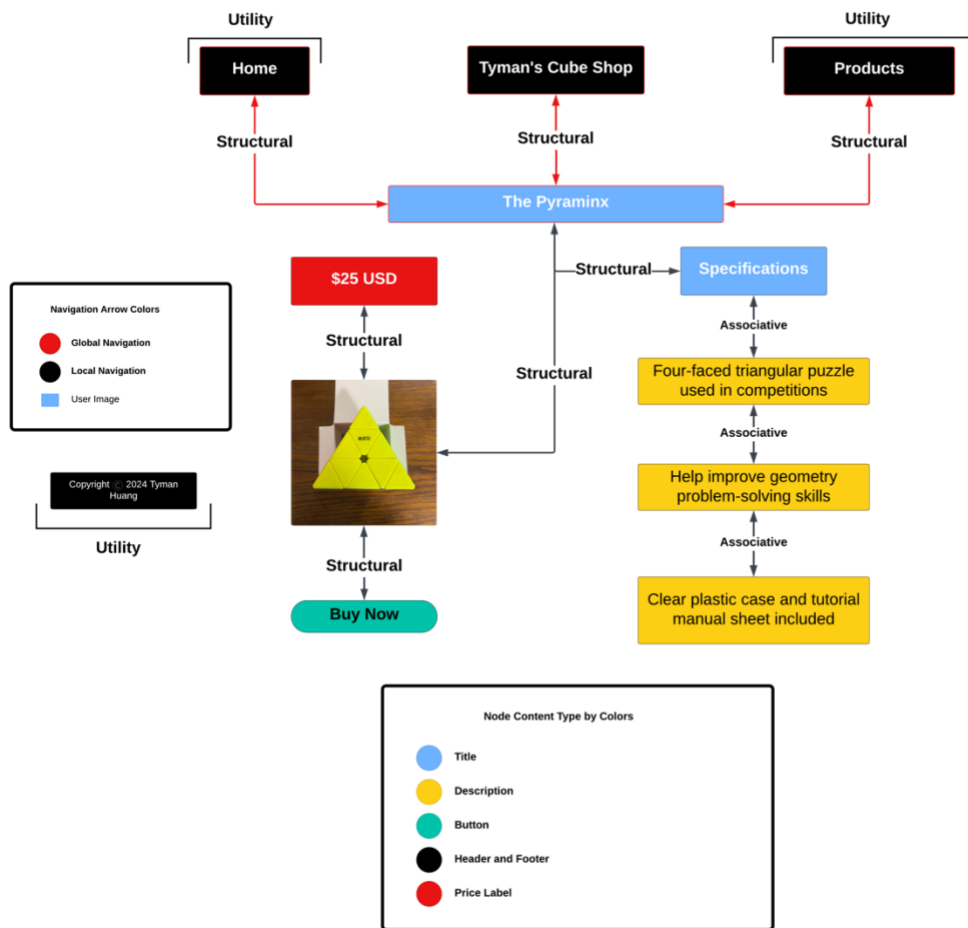
Information Architecture (IA) for Order Page (3x3 Rubik's Cube Example)



Information Architecture (IA) for Order Page (4x4 Rubik's Cube)



Information Architecture (IA) for Product Page (The Pyraminx)



Wireframes

The wireframes are produced in the Figma online tool, where each wireframe directly represents the final web pages. A prototype is then formed by adding animation connectors between the wireframes to illustrate the final product.

Prototype Link: <https://www.figma.com/proto/6kqCulxDyBJpDnmhsjmBa3/CCT360-Project-1?node-id=2-3&node-type=canvas&t=JU8JpupNIY6oMqLs-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=2%3A3>

Wireframe: Homepage



Wireframe: Products List

Home

Tyman’s Cube Shop

Products

All Rubik’s Cube Products

Square Layered Cubes

2x2 Cube

3x3 Cube

4x4 Cube

5x5 Cube

6x6 Cube

7x7 Cube

Special Polygon Cubes

The Pyraminx

The Skewb

The Megaminx

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Wireframes: Product Page

Home

Tyman’s Cube Shop

Products

GAN 13 Magnetic Rubik’s Cube



Buy Now \$20

Specifications

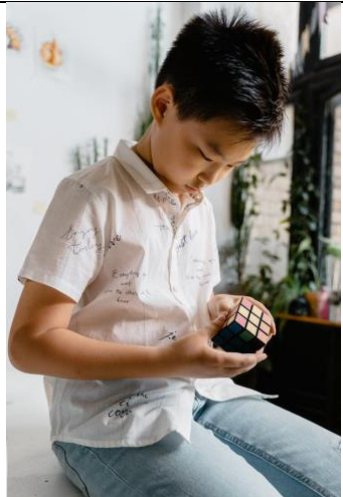
- Optimized for official Rubik’s Cube competitions worldwide
- Magnetic design allows smoother turns
- Storage box, mini screwdriver, and tutorial manual sheet included

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Wireframes: Order Page (Not Implemented in Final Website)

Order Checkout Form	
Gan 13 Magnetic Rubik's Cube 	Contact First Name Last Name Address City Postal Code State
Total \$20 USD	Payment Details Card Number Expiry Date CVV Code Order Your Cube Now!

Personas

	
Fig. 1. A person hovering a Rubik's cube, to illustrate sense of mastery (Persona 1)	Fig. 2. A child inspects the Rubik's cube with a confused look (Persona 2)

The first persona includes puzzle solvers with advanced experience who often attend Rubik's Cube competitions to build their portfolios and personal achievements. Concerning demographics, competitive solvers are mostly between 20 to 30 years old. Gender, ethnicity, income level, and education level are varied across these components,

as the two personas are distinct only by age distribution and skill level with solving different types of Rubik's Cubes. They devote their career and hobby to enjoying the lifestyle of solving twisty puzzles. Apart from competing with others, these adults are also customers of many Rubik's Cube stores since they often need replacements for new cubes. The primary goal they want to achieve is to find the most valued cubes on the market, which they desire for slightly lower prices but better quality. A challenge hindering their exploration to seek suitable low-priced puzzles is due to inflation, as the cost of everyday goods rises sharply, especially during the pandemic. It is frustrating if they have to pay a larger sum each time to buy the same bulk of puzzle toys. Lastly, the reasons for them to turn away from my shopping site are due to low stock of cubes and unfamiliarity with store quality and reputation.

The second persona is children and youth who have never interacted with the Rubik's Cube. For demographics, they range from 6 to 20 years old and are inexperienced with puzzle toys, whereas other traits can be varied. They are students in public or private schools, as playing the cubes is considered their hobby and integral to their cognitive intelligence development. If these children and youth are from higher-income families, finding cheap-priced cubes is only a secondary preference where they aim primarily for toy quality. One challenge children and youths face is purchasing the cubes without the allowance and permission from parents. Some parents support them as the cubes help develop muscle memory and logic skills, while others view them as distraction toys. The children and youth who acquire their first Rubik's Cube will be excited and curious about where the website description matches the product they receive. Finally, a crucial reason children and youths will leave my shop before ordering is having fewer photos and descriptions than competitor sites.

Usability Principles

Regarding website usability, the five principles to be evaluated with my Rubik's Cube shopping site are availability, clarity, recognition, credibility, and relevance (99 Designs Team, 2021). For availability, the website is only compatible with the desktop version and not yet developed for tablet and mobile device responsivity. Although the website has not reached the stage of optimization and advertising as an E-commerce site, the purpose of the website is intuitive, which aims to sell Rubik's Cubes at a value price. The page title and product cards on the landing page with prices and buy buttons are evident enough to be a shopping site. Next, my shopping site interface is recognizable and easily learned for adult and youth netizens, where color contrast, consistent font, and a simplistic layout demonstrate the website's aesthetics. However, credibility can be an issue when the site is

overly simplistic, where users typically expect more than 5 to 7 products for sale. The website is largely relevant to the two distinct personas discussed.

WCAG 2.0 Accessibility Principles

Perceivable: The design aesthetics of the page do not impair visibility, where color contrast is tested between foreground and background elements. Referencing from the color palette section, three color contrast checks are conducted. All the checks have contrast ratio scores beyond the WCAG minimum requirements, and the colors are not too similar for color blindness support. The font family is the same throughout the webpages to maintain consistency and all images contain alt texts. Labels and buttons are written to serve the intended usage after user interactions.

Operable: All page types of the shopping site are navigable between each other from buttons labeled with clear purposes. The destination of the buttons from the navigation bars of the home, product list, and product description page are consistent. Home and Product buttons on each bar end will return to the home and product list pages. The Shop Now button takes users to the product description page, while the Buy button takes them to the order page, which is highly intuitive.

Understandable: Each textual element throughout the website explains the exact purpose, where users are prompted to shop and buy the puzzle toys available. Simple English ensures that all webpage contents are easily understandable by youths and adults alike, where three bullet points are used to highlight each product's unique selling points.

Robust: The webpages are currently designed for desktop versions only where screen widths of 1000 to 2000 pixels should not overflow content. Mobile responsive versions will be further developed to be compatible with more device types, as a weakness of my current shopping site. All elements are spoken from top to bottom and left to right when using a screen reader, increasing the robustness for users requiring audio.

References

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